

Introduction to Nonparametric Statistics

TD 3: Robust Tests

Exercise 1: A criticism often directed at the film industry is its preference for young actresses, whereas male actors of all ages can obtain major roles. To test this hypothesis, we record the ages of the male and female leads in the 9 films nominated for the 2014 Academy Award for Best Picture.

Age of the leading actor	40	57	44	52	39.2	77	48	36	39
Age of the leading actress	39.5	55	41.5	49	39.5	84	79	30	23

Table 1: In order: *American Hustle*, *Captain Phillips*, *Dallas Buyers Club*, *Gravity*, *Her*, *The Wolf of Wall Street*, *Nebraska*, *Philomena*, *12 Years a Slave*

We want to know whether actresses are younger than actors in successful American movies. We propose to use the Wilcoxon signed-rank test. We treat age as a continuous variable.

1. Explain why we use the Wilcoxon signed-rank test here rather than the sign test.
2. State the null hypothesis and the alternative hypothesis. Recall the expression of the test statistic W_9^+ and its expectation under H_0 .
3. Compute the p -value of the test using the table below, which corresponds to the distribution of W_9^+ under H_0 .
4. Conclude for a test at level $\alpha = 5\%$.

k	0	1	2	3	4	5	6	7	8	9
$P[W_9^+ \leq k]$	0.0019	0.0039	0.0058	0.0097	0.0136	0.0195	0.0273	0.0371	0.0488	0.0644
k	10	11	12	13	14	15	16	17	18	19
$P[W_9^+ \leq k]$	0.0820	0.1015	0.125	0.1503	0.1796	0.2128	0.2480	0.2851	0.3261	0.3671
k	20	21	22	23	24	25	26	27	28	29
$P[W_9^+ \leq k]$	0.4101	0.4550	0.5	0.5449	0.5898	0.6328	0.6738	0.7148	0.7519	0.7871
k	30	31	32	33	34	35	36	37	38	39
$P[W_9^+ \leq k]$	0.8203	0.8496	0.875	0.8984	0.9179	0.9355	0.9511	0.9628	0.9726	0.9804
k	40	41	42	43	44	45				
$P[w_9^+ \leq k]$	0.9863	0.9902	0.9941	0.9960	0.9980	1				

Exercise 2: We study the effect of a low dose of Cambendazole on *Trichinella spiralis* infections in mice. Sixteen mice were infected with the same number of *Trichinella* larvae and were then randomly split into two groups. The first group of 7 mice received Cambendazole, at a dose of 10 mg per kilogram, 60 hours after infection. The other 9 mice received no treatment. After one week, all mice were sacrificed and the following numbers of adult worms were recovered from the intestines:

Untreated mice	51	55	62	63	65	68	71	75	79
Treated mice	47	49	53	57	60	61	67		

Describe a statistical protocol to test a possible efficacy of Cambendazole for treating *Trichinella spiralis* infections in mice. Using only the Mann–Whitney table given below, for what test level α can we be sure that the Mann–Whitney test will conclude that Cambendazole is effective for treating *Trichinella spiralis* infections in mice?

We give a few values of the cdf $F_{n,p}$ of the Mann–Whitney statistic under H_0 for $n = 9$ and $p = 7$: $F_{9,7}(10) = 0.011$, $F_{9,7}(11) = 0.016$, $F_{9,7}(12) = 0.021$.

Exercise 3: We want to test the effectiveness of a new migraine treatment. We have a sample of 18 people who suffer from migraines, and we provide each of them with an equal number of pills corresponding to the new treatment (A) and to standard aspirin (B). After they have used both sets of pills, each patient is asked to judge which type of pill (A or B) was more effective. Among the 18 patients, 12 report that the new treatment (A) is more effective than the old one (B). How can we test the effectiveness of the new treatment?